

CS & IT ENGINEERING

DISCRETE MATHS
SET THEORY



Lecture No. 15



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TOPICS TO BE COVERED

01 Algebraic structures

02 Semi Group

03 Monoid

04 Group

GROUP THEORY



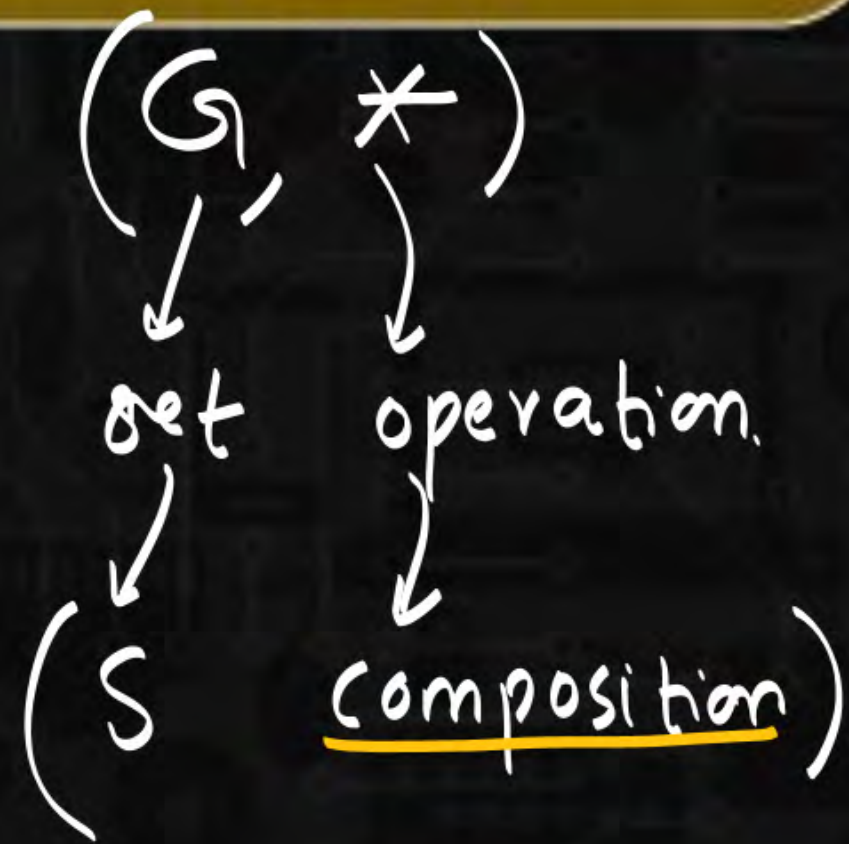
$A = \{1, 2, 3\}$ operation: composition of function.

$1!1!1!$ $3! = 6$

$$S = \left\{ \begin{array}{l} f_1: \begin{array}{l} 1 \rightarrow 1 \\ 2 \rightarrow 2 \\ 3 \rightarrow 3 \end{array}, \quad f_2: \begin{array}{l} 1 \rightarrow 2 \\ 2 \rightarrow 1 \\ 3 \rightarrow 3 \end{array}, \quad f_3: \begin{array}{l} 1 \rightarrow 3 \\ 2 \rightarrow 1 \\ 3 \rightarrow 2 \end{array}, \quad f_4: \begin{array}{l} 1 \rightarrow 3 \\ 2 \rightarrow 2 \\ 3 \rightarrow 1 \end{array}, \quad f_5: \begin{array}{l} 1 \rightarrow 1 \\ 2 \rightarrow 3 \\ 3 \rightarrow 2 \end{array}, \quad f_6: \begin{array}{l} 1 \rightarrow 2 \\ 2 \rightarrow 3 \\ 3 \rightarrow 1 \end{array} \end{array} \right\}$$

$(G, *)$

GROUP THEORY



$$S = \{ f_1, f_2, f_3, f_4, f_5, f_6 \}$$

GROUP THEORY



$$A = \{1, 2, 3\}$$

operation: composition of function.

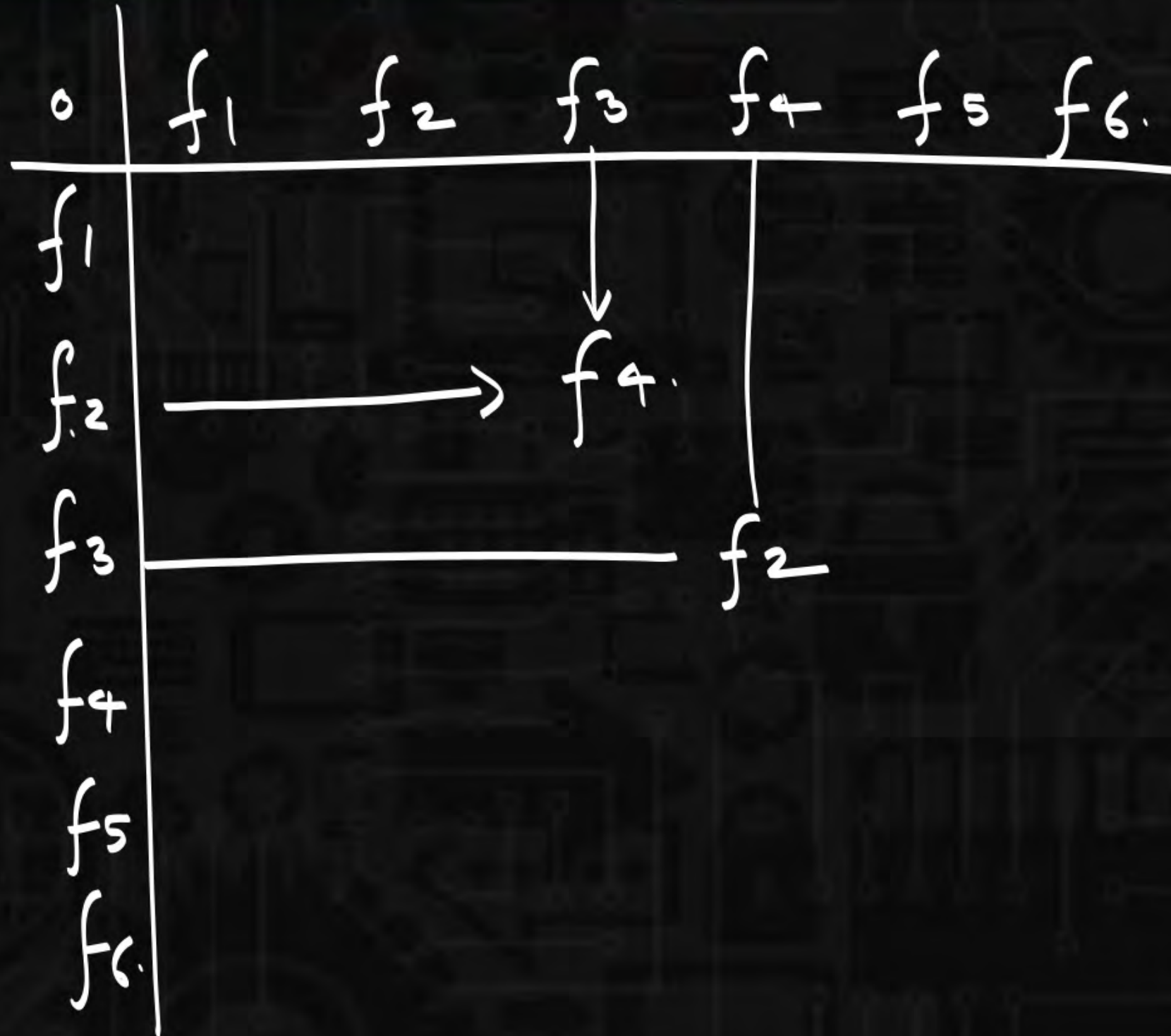
$S = \{ \underbrace{1, 1, 1}_e, 3! = 6 \}$

$f_1: \begin{matrix} 1 \rightarrow 1 \\ 2 \rightarrow 2 \\ 3 \rightarrow 3 \end{matrix}$	$f_2: \begin{matrix} 1 \rightarrow 2 \\ 2 \rightarrow 1 \\ 3 \rightarrow 3 \end{matrix}$	$f_3: \begin{matrix} 1 \rightarrow 3 \\ 2 \rightarrow 1 \\ 3 \rightarrow 2 \end{matrix}$	$\underline{\underline{f_4: \begin{matrix} 1 \rightarrow 3 \\ 2 \rightarrow 2 \\ 3 \rightarrow 1 \end{matrix}}}$	$f_5: \begin{matrix} 1 \rightarrow 1 \\ 2 \rightarrow 3 \\ 3 \rightarrow 2 \end{matrix}$	$f_6: \begin{matrix} 1 \rightarrow 2 \\ 2 \rightarrow 3 \\ 3 \rightarrow 1 \end{matrix}$
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$\underline{\underline{f_2 \circ f_3}} = f_2 f_3 = f_4$

$\underline{\underline{f_3 \circ f_4}} = \begin{matrix} 1 \rightarrow 2 \\ 2 \rightarrow 1 \\ 3 \rightarrow 3 \end{matrix} f_2$

GROUP THEORY



GROUP THEORY



$$f_1: \begin{array}{l} 1-1 \\ 2-2 \\ 3-3 \end{array}$$

$$f_2: \begin{array}{l} 1-2 \\ 2-1 \\ 3-3 \end{array}$$

$$f_3: \begin{array}{l} 1-3 \\ 2-1 \\ 3-2 \end{array}$$

$$f_4: \begin{array}{l} 1-3 \\ 2-2 \\ 3-1 \end{array}$$

$$f_5: \begin{array}{l} 1-1 \\ 2-3 \\ 3-2 \end{array}$$

$$f_6: \begin{array}{l} 1-2 \\ 2-3 \\ 3-1 \end{array}$$

$$(f_2 \circ f_3) \circ f_4 = f_2 \circ (f_3 \circ f_4)$$

$$\frac{f_4 \circ f_4}{f_1}$$

$$\begin{array}{l} 1-3-2 \\ 2-2-1 \\ 3-1-3 \end{array} = \begin{array}{l} 1-2 \\ 2-1 \\ 3-3 \end{array}$$

$$\frac{f_2 \circ f_2}{f_1}$$

$$\begin{array}{l} 1-2-1 \\ 2-1-2 \\ 3-3-3 \end{array} = f_1$$

$$\frac{f_2 \circ f_3}{f_3 \quad f_2} = \frac{1-3-3}{2-1-2} = \frac{1-3-1}{2-2-2} = f_1$$

$$\frac{3-2-1}{3-1-3} = f_4$$

$$f_4 \circ f_4$$

$$f_4 \circ f_4 = f_1$$

$$f_2 \circ f_2 = f_1$$

GROUP THEORY



1) Closed. ✓

2) Associative. ✓

3) identity (f_1) ✓

4) Inverse ✓

(S, o) Group. ✓

$$f_2 \circ f_2 = f_1$$

$$f_4 \circ f_4 = f_1$$

$$\underline{f_3' = f_6}$$

$$f_5' =$$

$$f_6' =$$

$$f_2' = f_2$$

$$f_4' = f_4$$

GROUP THEORY

$\oplus_6$	0	1	2	3	4	5
0	0	1	2	3	4	5
1	1	2	3	4	5	0
2	2	3	4	5	0	1
3	3	4	5	0	1	2
4	4	5	0	1	2	3
5	5	0	1	2	3	4

Subgroup (H)

H is called sub/group.

1. $H \subseteq G$.

2. { Closed
Associative
identity.
Inverse.

1. $H \subseteq G$.

2. H should also be Group.

GROUP THEORY

$\oplus_6$	0	1	2	3	4	5
0	0	1	2	3	4	5
1	1	2	3	4	5	0
2	2	3	4	5	0	1
3	3	4	5	0	1	2
4	4	5	0	1	2	3
5	5	0	1	2	3	4

$$H_1 = \{1, 3, 5\}$$

Check H_1 is subgroup or not?

1) $H_1 \subseteq G$ $\{1, 3, 5\} \subseteq \{0, 1, 2, 3, 4, 5\}$

2) $\rightarrow$ closed.

H_1 is not subgroup

$\oplus_6$	1	3	5
1	2		
3			
5			

GROUP THEORY



$$G = \{0, 1, 2, 3, 4, 5\}$$

Closed.

Associative

identity

Inverse

0

unique.

$$H_1 = \{1, 3, 5\}$$

0 is not here.

1) $H_1 \subseteq G$.

2)

C

A

I

In.

unique

GROUP THEORY



$\oplus_6$	0	1	2	3	4	5
0	0	1	2	3	4	5
1	1	2	3	4	5	0
2	2	3	4	5	0	1
3	3	4	5	0	1	2
4	4	5	0	1	2	3
5	5	0	1	2	3	4

$$H_2 = \{0, 1, 3\}$$

1) $H_2 \subseteq G$.

2) a) closed.

H_2 is not subgroup.

	0	1	3
0	0	1	3
1	1	2	4
3	3	4	2

not closed.

GROUP THEORY



$\oplus_6$	0	1	2	3	4	5
0	0	1	2	3	4	5
1	1	2	3	4	5	0
2	2	3	4	5	0	1
3	3	4	5	0	1	2
4	4	5	0	1	2	3
5	5	0	1	2	3	4

* $H_3 = \{0, 2, 4\}$

1) $H_3 \subseteq G$

2) a) closed. ✓

b) Associative

3) identity.

4) Inverse.

each row $\rightarrow 0$

	0	2	4
0	0	2	4
2	2	4	0
4	4	0	2

$0' = 0$
 $2' = 4$
 $4' = 2$

GROUP THEORY



Lagrange's thm:

if H is subgroup of G then $\frac{|G|}{|H|}$

$$H_3 = \{0, 2, 4\} \quad G = \{0, 1, 2, 3, 4, 5\}$$

$$|H_3| = 3.$$

$$|G| = 6.$$

$$\text{if } H_3 \text{ is subgroup of } G. \quad \frac{6}{3}$$

viceversa is not true

$$H_2 = \{1, 3, 5\} \quad |H_2| = 3.$$

$$|G| = 6.$$

GROUP THEORY



Every Group contains 2 subgroups: (Trivial)

$$1) \quad e \subseteq G. \quad \begin{array}{c} e \\ e | e \end{array}$$

$$2) \quad G \subseteq G.$$

1) closed ✓

2) Associative. ✓

$$e * (e * e) = (e * e) * e.$$

$$3) \quad e * e = e$$

$$4) \quad e * e = e.$$

GROUP THEORY



$$H > \underline{66} \text{ \& } \div \underline{6}$$

$$K \subset H \subset G$$

$$|K| = 66$$

$$|G| = 660$$

$$K \subset \underline{66}$$

$$11 \times 6$$

$$H \subset G$$

$$\downarrow$$

$$66 \times 2 = \underline{132}$$

OR

$$66 \times 5 = \underline{330}$$

$$G \downarrow$$

$$660$$

$$66 \times 10$$

$$66 \times 2 \times 5$$

GROUP THEORY



$$|G| = 84.$$

if H is subgroup of G . what will largest size of H .
 $|H| = 84.$

2) $H \subset G$. $|H| = 42$.
what will be largest size of H .

$$\{1, \dots, \textcircled{42}, \textcircled{84}\}$$

GROUP THEORY

Exponential of elements:

$$a^1 = a.$$

$$a^2 = a * a =$$

$$a^3 = a * a * a =$$

$$\underline{\langle a \rangle} = \{ a, a^2, a^3, \dots \}$$

GROUP THEORY

$\oplus_6$	0	1	2	3	4	5
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1	1	2	3	4	5	0
2	2	3	4	5	0	1
3	3	4	5	0	1	2
4	4	5	0	1	2	3
5	5	0	1	2	3	4

$$3^1 = \underline{\underline{3}}$$

$$3^2 = 3 \oplus_6 3 = 0$$

$$3^3 = 0 \oplus_6 3 = \underline{\underline{3}}$$

$$\underline{\underline{\langle 3 \rangle = \{0, 3\}}}$$

3 has generated $\{0, 3\}$

GROUP THEORY



$\oplus_6$	0	1	2	3	4	5
0	0	1	2	3	4	5
1	1	2	3	4	5	0
2	2	3	4	5	0	1
3	3	4	5	0	1	2
4	4	5	0	1	2	3
5	5	0	1	2	3	4

$$2^1 = 2$$

$$2^2 = 2 * 2 = 4$$

$$2^3 = 4 * 2 = 0$$

$$2^4 = 0 * 2 = 2$$

$$\langle 2 \rangle = \{0, 2, 4\}$$

$$4^1 = 4$$

$$4^2 = 4 * 4 = 2$$

$$4^3 = 2 * 4 = 0$$

$$4^4 = 0 * 4 = 4$$



$$\langle 4 \rangle = \{0, 2, 4\}$$

GROUP THEORY



$\oplus_6$	0	1	2	3	4	5
0	0	1	2	3	4	5
1	1	2	3	4	5	0
2	2	3	4	5	0	1
3	3	4	5	0	1	2
4	4	5	0	1	2	3
5	5	0	1	2	3	4

$$1^1 = 1 \quad 1^2 = 2 \quad 1^3 = 2 * 1 = 3$$

$$1^4 = 3 * 1 = 4 \quad 1^5 = 4 * 1 = 5$$

$$1^6 = 5 * 1 = 0 \quad 1^7 = 0 * 1 = 1$$

$$\langle 1 \rangle = \{0, 1, 2, 3, 4, 5\}$$

1 has generated all elements in the group.

1 is called generator.

a has generated all elements in the group.
 a is called generator.

Group contains generator.

Group is called cyclic group.

→ Subgroup of cyclic group is also cyclic group.

GROUP THEORY

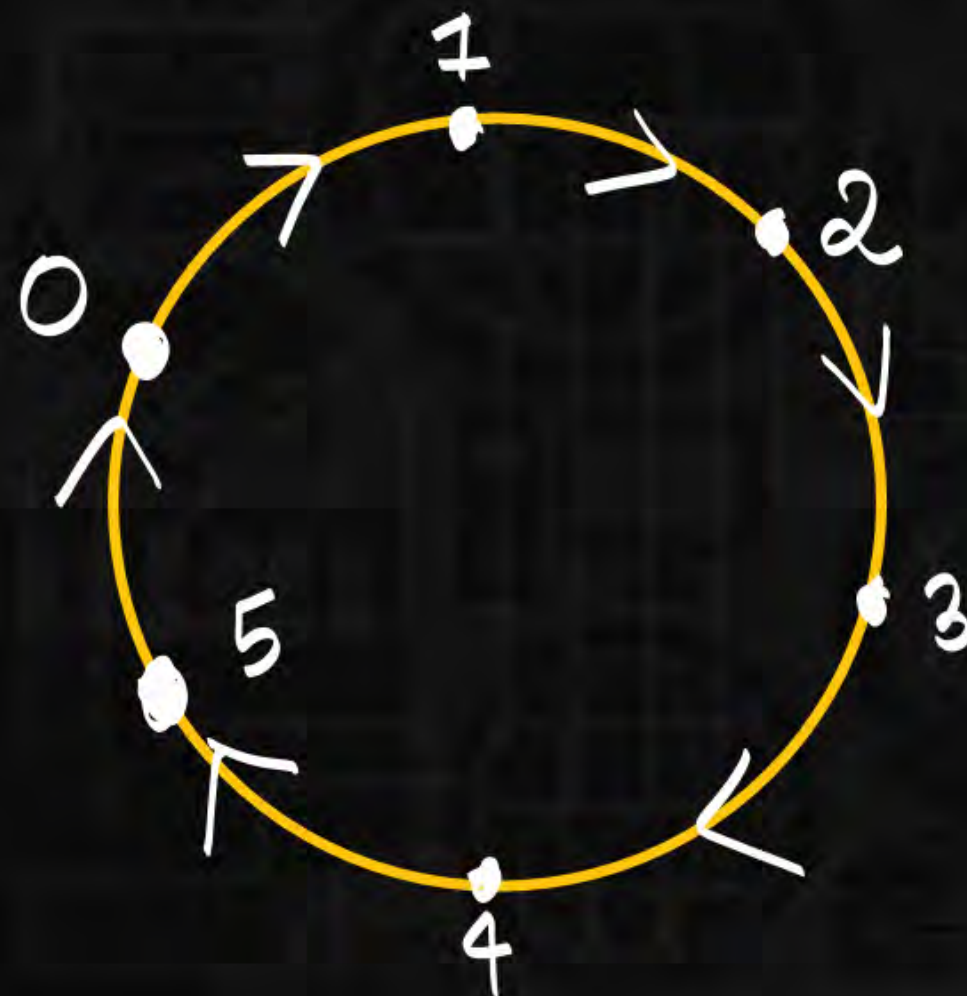
$\oplus_6$	0	1	2	3	4	5
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2	2	3	4	5	0	1
3	3	4	5	0	1	2
4	4	5	0	1	2	3
5	5	0	1	2	3	4

$$1^1 = 1 \quad 1^2 = 2 \quad 1^3 = 2 * 1 = 3$$

$$1^4 = 3 * 1 = 4 \quad 1^5 = 4 * 1 = 5$$

$$1^6 = 5 * 1 = 0 \quad 1^7 = 0 * 1 = 1$$

$$\langle 1 \rangle = \{0, 1, 2, 3, 4, 5\}$$



GROUP THEORY



$\oplus_6$	0	1	2	3	4	5
0	0	1	2	3	4	5
1	1	2	3	4	5	0
2	2	3	4	5	0	1
3	3	4	5	0	1	2
4	4	5	0	1	2	3
5	5	0	1	2	3	4

$$5^1 = 5$$

$$5^2 = 5 * 5 = 4$$

$$5^3 = 4 * 5 = 3$$

$$5^4 = 3 * 5 = 2$$

$$5^5 = 2 * 5 = 1$$

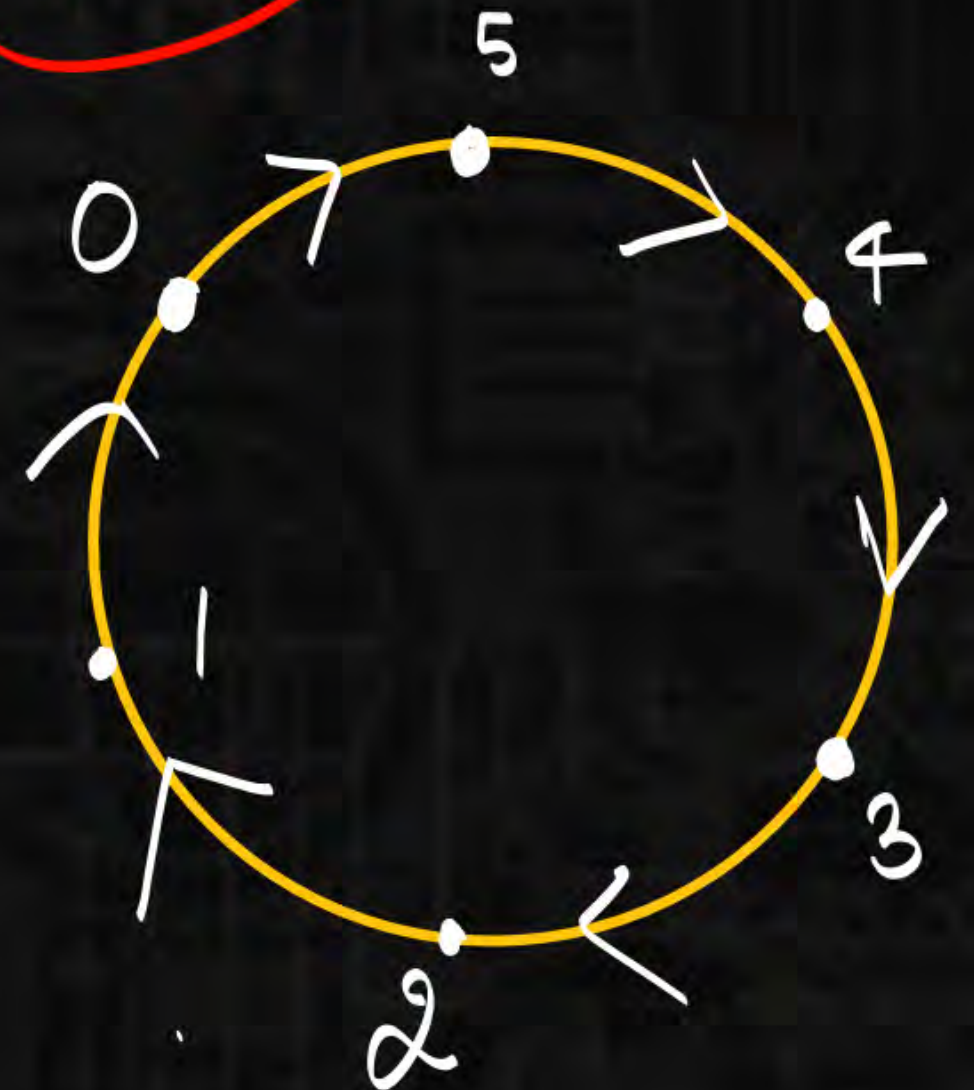
$$5^6 = 1 * 5 = 0$$

$$5^7 = 0 * 5 = 5$$

$$\langle 5 \rangle$$

$$= \{0, 1, 2, 3, 4, 5\}$$

$$1^1 = 5$$



GROUP THEORY



Closed

Associative

identity

Inverse.

Semigroup

monoid

Group.

GROUP THEORY



Closed	Associative	Identity	Inverse	Commutative
semigroup.				
monoid.				
Group.				
Abelian Group.				

GROUP THEORY



$$a = e$$

$$\{1, 2, 4, 7, 8, 11, 13, 14\}$$

is group under multiplication, mod 15.

Inverse of 4 & 7 (GATE -05)

- a) 3, 13 c) 4, 13
b) 2, 11 d) 8, 14.

GROUP THEORY



$$|G| = 15$$

L be subgroup G .

$$L \neq G.$$

size of L is at least 4.

$$|L| = \underline{\quad}. \quad (\text{GATE-14})$$

GROUP THEORY



	e	a	b	c
e	e	a	b	c
a	a	b	c	e
b	—	—	—	—
c	—	—	—	—

last row?

a) caeb

b) cbae

c) cbea

d) ceab

GROUP THEORY



	a	b	c	d
a	a	b	c	d
b	b	a	d	c
c	c	d	b	a
d	d	c	a	b

Generator?

